

Cassandra Aggregation Example Scenario

Scenario: Employee Salary and Age Analysis

The company has a table `employees` with the following schema:

```
CREATE TABLE employees (  
    employee_id UUID PRIMARY KEY,  
    name text,  
    age int,  
    department text,  
    salary decimal  
);
```

Business Requirements:

1. Total Number of Employees: The company wants to know the total number of employees.
2. Total Salary in Each Department: Calculate the total salary expenditure for each department.
3. Average Salary in Each Department: Find the average salary of employees in each department.
4. Oldest and Youngest Employees: Identify the oldest and youngest employees in the company.
5. Grouped Salary and Age Information: Group the salary and age information by department.
6. Order Employees by Age: Order the employees by age in descending order.
7. Filter Employees: Allow filtering to find specific subsets of data.

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Example Queries Using All Aggregation Methods

-- 1. Total Number of Employees

```
SELECT COUNT(*) FROM employees;
```

-- 2. Total Salary in Each Department

```
SELECT department, SUM(salary) FROM employees GROUP BY department;
```

-- 3. Average Salary in Each Department

```
SELECT department, AVG(salary) FROM employees GROUP BY department;
```

-- 4. Oldest and Youngest Employees

```
SELECT MAX(age) FROM employees;
```

```
SELECT MIN(age) FROM employees;
```

-- 5. Grouped Salary and Age Information

```
SELECT department, COUNT(*), SUM(salary), AVG(salary), MAX(age), MIN(age)  
FROM employees  
GROUP BY department;
```

-- 6. Order Employees by Age

```
SELECT * FROM employees ORDER BY age DESC ALLOW FILTERING;
```

-- 7. Filter Employees (with caution)

```
SELECT * FROM employees WHERE age > 30 ALLOW FILTERING;
```